National University of Computer & Emerging Sciences, Karachi

Spring-2018 CS-Department

MidTerm 2

4th April 2018, 09:00 am – 10:00 am

Course Code: CS402	**Course Name:** Compiler Construction
Instructor Name / Names: M. Shahzad	

Instructions:

- Return the question paper.
- Read each question completely before answering it. There are **3 questions and 2 pages.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. All the questions should be attempted on the answer sheet.
- All questions carry equal marks and equally distributed in sub parts.

Time: 60 minutes. **Max Marks:** 30 points

Question 1 (10 points = 2 * 5):

Provide 2-3 line replies to all of the following short questions. Answer that exceeds 3 lines will not be considered.

1) Describe the relationship between a production and an item in an LR(0) grammar. [2 points]
2) What are the error handling mechanisms in parsing phase? [2 points]
3) Demonstrate stack implementation in implementation of shift reduce Parsing. [2 points]
4) What are the requirements of the grammar to be in LL(1) grammar? [2 points]
5) Explain the limitation of the recursive descent parsing. [2 points]

Question 2 (10 points = 2 * 5):

Consider the following grammar H (the different productions have been numbered):

1. S → [SX]
2. S → a
3. X → ε
4. X → +SY
5. X → Yb
6. Y → ε
7. Y → -SXc

It contains 3 non-terminals S, X and Y, along with terminals [,], a, b, c, +, - and ε.

a) Fill in the table below with the First and Follow sets for the non-terminals in this grammar [5 points]

	FIRST	FOLLOW
S	(, a	
X	+, -, b, ε	
Y	-, ε	

b) Fill in the parse table below for the grammar H (don't write the productions inside the table; write only their numbers as given above) [5 points]

	a	b	c	+	-	[	]	$
S								
X								
Y								

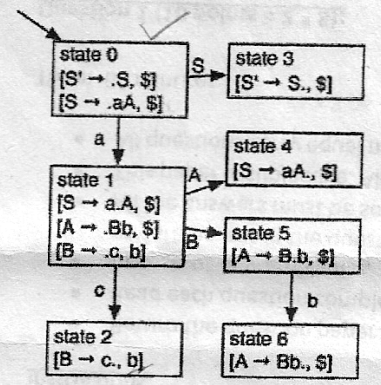
Question 3 (10 points = 2* 5):

a) Draw the LR(1) parsing DFA for the following grammar. [5 points]

1) S -> aSb

2) S ->c

b) Write down the grammar and the **ACTION** and **GOTO** tables corresponding to the following LR(1) parsing DFA. [5 points]



S → aA
A → Bb
B → c

BEST OF LUCK!